KOÇ UNIVERSITY	ELEC 205 Assignment 3 Sequential Circuit Design
	Deadline: May 11, 11:59 PM
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Task 1

I draw the diagram in Figure 1 with 5 states (A/00, B/00, C/00, D/10, E/01). In these states the most significant bit shows Z_1, and the least significant bit shows Z_0.

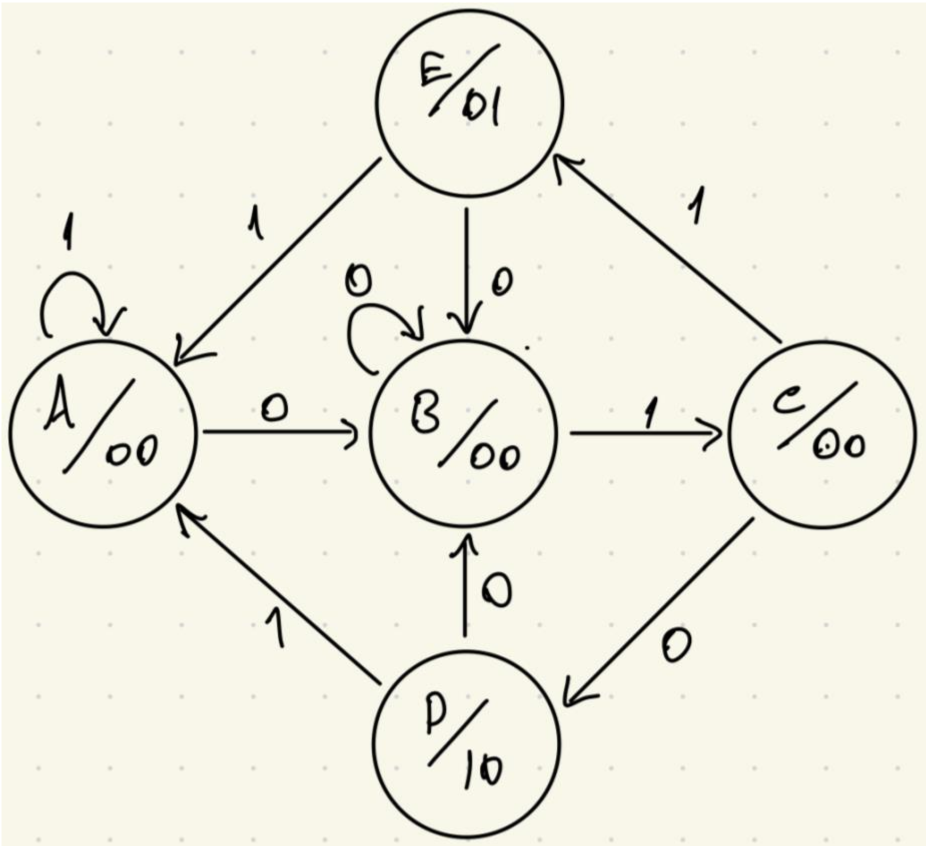

Figure 1

Task 2

Because we have 5 states, we need at least 3 binary digits to represent states. I represented the states with the binary codes in Table 1. Unused binary codes are treated as don't care condition (x).

State	Binary Code ($S_2S_1S_0$)
A	000
B	001
C	010
D	011
E	100

Table 1

Task 3

The state transition table is in Table 2 with unused state binary codes as don't care conditions.

Present State ($S_2S_1S_0$)	**Input (X)**	**Next State**	**Output (Z_1Z_0)**
A (000)	0	B (001)	00
A (000)	1	A (000)	00
B (001)	0	B (001)	00
B (001)	1	C (010)	00
C (010)	0	D (011)	00
C (010)	1	E (100)	00
D (011)	0	B (001)	10
D (011)	1	A (000)	10
E (100)	0	B (001)	01
E (100)	1	A (000)	01
x (101)	x	x	xx
x (110)	x	x	xx
x (111)	x	x	xx

Table 2

Task 4

To see Boolean expressions of state binary codes and outputs explicitly, I created K-Maps below.

- $S_0^+ = \sum m(0,2,4,6,8),\ \sum d(10,11,12,13,14,15)$

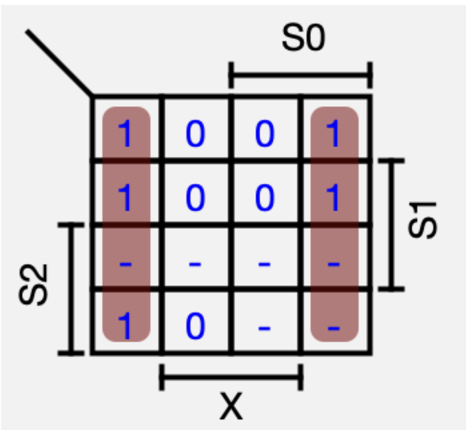

Figure 2

  - $S_0^+ = X'$

- $S_1^+ = \sum m(3,4),\ \sum d(10,11,12,13,14,15)$

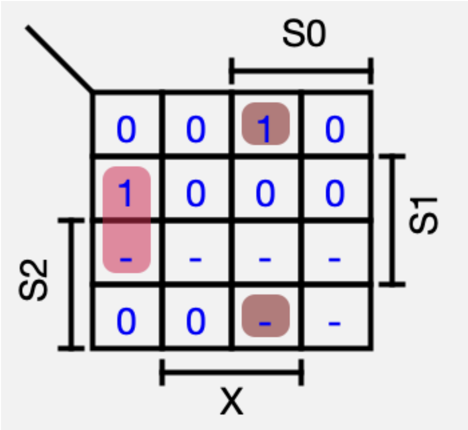

Figure 3

  - $S_1^+ = S_0'S_1X' + S_0S_1'X$ = NAND (NAND (S_0', S_1, X'), NAND (S_0, S_1', X))

- $S_2^+ = \sum m(5),\ \sum d(10,11,12,13,14,15)$

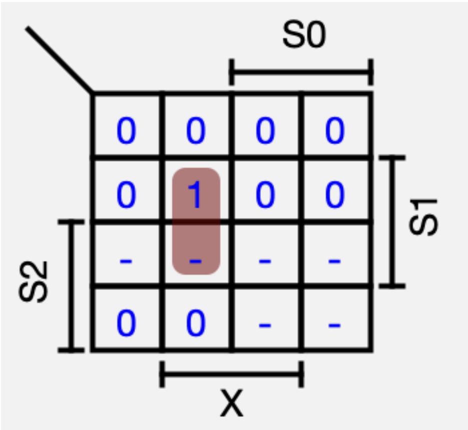

Figure 4

  - $S_2^+ = S_0'S_1X$ = NAND (NAND (S_0', S_1, X), NAND (S_0', S_1, X))

- $Z_0 = \sum m(8,9),\ \sum d(10,11,12,13,14,15)$

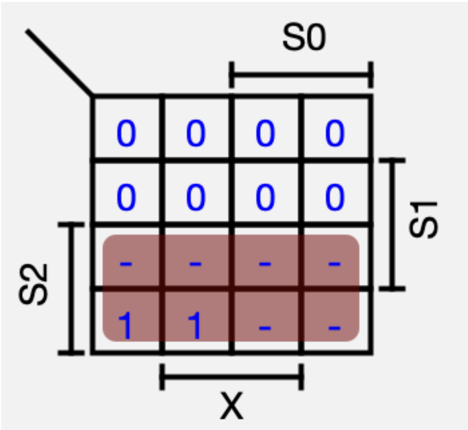

Figure 5

  - $Z_0 = S_2$

- $Z_1 = \sum m(6,7),\ \sum d(10,11,12,13,14,15)$

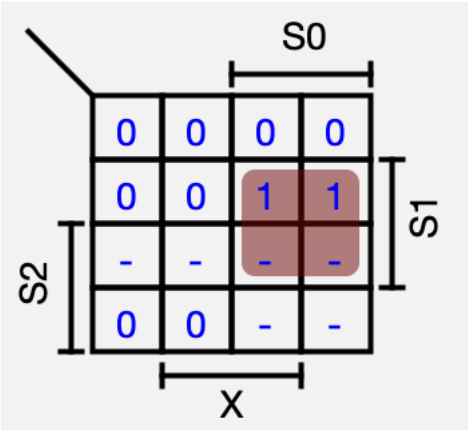

Figure 6

  - $Z_1 = S_0S_1$ = NAND (NAND (S_0, S_1), NAND (S_0, S_1))

These K-Maps help us to generate explicit Boolean expressions of state binary codes and outputs using only NAND gates. Now we can draw the circuit in Task 5 using D flip-flops to represent each binary code of states and NAND gates as the expressions that we found in this task.

Task 5

I created the circuit in Figure 7 according to expression which I found in Task 4.

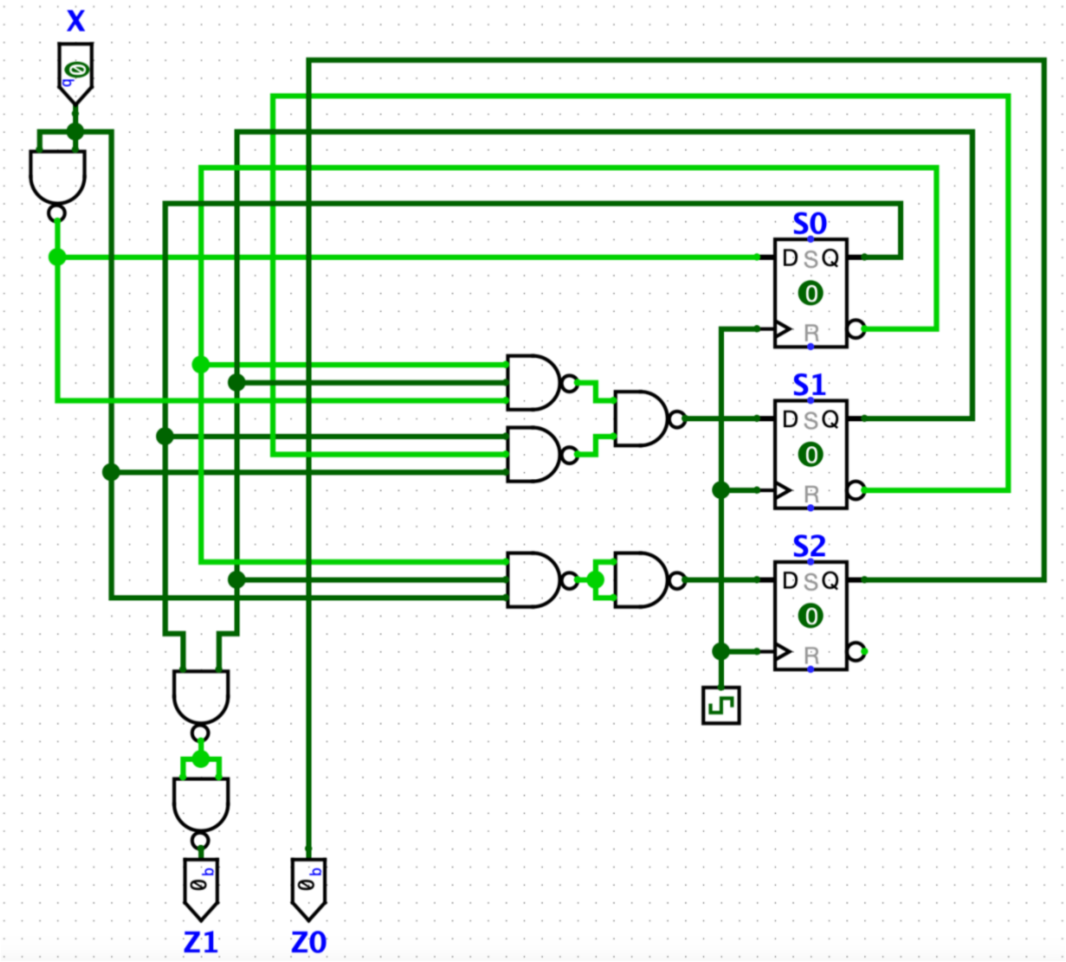

Figure 7

Task 6

The timing diagram contains 20 consecutive clock cycles with 3 (0 →1 →0) patterns and 3 (0 →1 →1) patterns as you can see in the diagram. Because of the structure of the D flip-flop in Logisim, which is Edge-Triggered, our states' values changes only when clock is 1 since slave of the D flip-flop is clock value 1. At the same time master of the D flip-flop is clock value 0, so when we determine the pattern, we have to check the X values when the clock is 0.

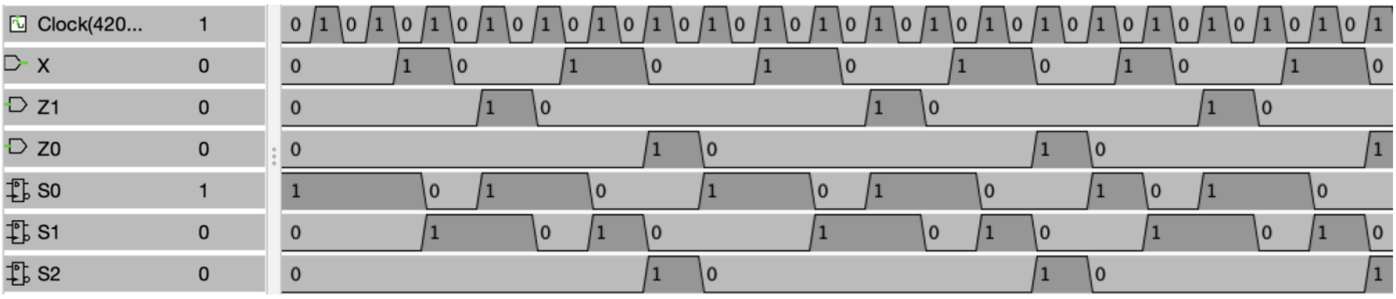

Figure 8

When we check the timing diagram in Figure 8 according to my explanation in previous paragraph, output of the timing diagram seems correct.